Kingdom of Saudi Arabia
Ministry of Education
Taibah University
College of Computer Science and Engineering
Department of Artificial Intelligence and Data Science

المملكة العربية السعودية
وزارة التعليم
جامعة طيبة
كلية علوم وهندسة الحاسب الآلي
قسم الذكاء الاصطناعي وعلوم البيانات

AI352 Course Project

Natural Language Processing

Arabic Sentiment Classification using Traditional NLP and LLM-Based Approaches

Section: F03

Prepared by

Bushra Abdulrahman Almutrafi	4456347
Shumukh Eid Altoom	4451941
Layal Mohammed Alhazmi	4451274

Supervised by

Dr. Ghada Alharbi

Spring 2026

Contents

1 Abstract

This project focuses on emotion classification using Arabic texts, by applying supervised machine learning techniques such as Random Forest, SVM, and logistic regression, with an emphasis on language-specific preprocessing and model behavior analysis using different feature engineering methods. Additionally, a large language model (LLM) approach was used, and a comparison between the two approaches was made.

Arabic natural language processing presents additional challenges compared to English, including complex morphological structure, dialectal diversity, segmentation difficulties, and the lack of standardized processing pipelines. This project explores how traditional natural language processing methods and large language model (LLM) approaches perform under these constraints.

2 Introduction

Recently, Large Language Models (LLMs) have become dominant in the field of Artificial Intelligence (AI), as large companies have begun to launch new models continuously and compete to build systems that are more powerful and capable of understanding human language. Despite this, traditional approaches of Natural Language Processing (NLP) and Machine Learning (ML) are still widely used due to their low computational cost and ease of interpreting their results. In this Project, we focus on discovering emotions in Arabic tweets. In addition, we will compare traditional approaches with Large Language Models (LLMs) approaches to evaluate their performance in understanding emotions in Arabic texts.

3 Problem Definition

Despite many modern technologies, the Arabic language still poses a challenge in the field of Natural Language Processing (NLP) due to the complexity of its structure and the abundance of its dialects. In this project, we made sure that the data used in the project was tweets in Arabic to contain several dialects and in an unorganized written format. We will see whether the Large Language Models (LLMs) will excel or whether there are still limits in performance.

4 Corpus Description

The original Arabic tweet dataset was reduced into a binary sentiment classification problem by mapping emotions into Positive and Negative classes and removing none and surprise labels. So the final dataset contained 7,469 tweets.

Feature	**Description**
Original dataset size	10,065 records and 3 columns
Reduced dataset size	7,469 records
Columns	ID, TWEET, LABEL
ID	Numeric identifier for each tweet
TWEET	Text column containing the tweet content
Original LABEL	joy, love, sympathy, anger, sadness, fear, none, and surprise
Reduced LABEL	Positive and Negative
Label mapping	joy, love, and sympathy were mapped to Positive. Anger, sadness, and fear were mapped to Negative. Tweets labeled none and surprise were removed from the dataset.
Missing values in original dataset	One missing value in the TWEET column only
Average text length in original dataset	About 14.5 words
Minimum text length in original dataset	1 word
Maximum text length in original dataset	35 words
Class distribution after reduction	Negative: 3907, Positive: 3562

Table 1: Dataset Description

5 Related Work

A previous study by Al-Khatib and El-Beltagy used the same Arabic tweet dataset used in this project to detect emotions. They applied traditional machine learning methods such as Naïve Bayes, Complement Naïve Bayes, and Sequential Minimal Optimization. They achieved the best result using Complement Naïve Bayes with an overall accuracy of 68.12%. However, the study only focused on traditional approach. Therefore, our project aims to compare both Large Language Models (LLMs) and Traditional approaches [1].

6 Methodology

6.1 Traditional Approach

The traditional or classical approach to solving NLP is a sequential flow of several key steps, and it is a statistical approach. When we take a closer look at a traditional NLP learning model, we will be able to see a set of distinct tasks taking place, such as preprocessing data by removing unwanted data, feature engineering to get good numerical representations of textual data, learning to use machine learning algorithms with the aid of training data, and predicting outputs for novel unfamiliar data. Of these, feature engineering was the most time-consuming and crucial step for obtaining good performance on a given NLP task [2].

6.1.1 Data Preprocessing

As we explored the data, we found that there were 44 duplicate tweets, 8 of which contained more than one label, and the rest shared the same label. We found that their presence would not affect. So we did not address it. We also found that there was one missing value in the tweet column, and we deleted it. We also made sure that there were no hashtags, mentions, or links. Then we started normalizing the tweet column because we want to unify the text so that the subsequent steps work properly. We dealt with the Diacritics and deleted them, in addition to punctuation marks and strange symbols, including emojis, stop words, and English words, since the project aims to analyze emotions from Arabic tweets only. then tokenization was applied to split each tweet into individual words, allowing lemmatization to be performed at the word level. After applying these preprocessing steps, the tokens were joined back into text format for feature extraction. Finally, before The dataset was split into 70% training, 15% validation, and 15% testing sets using stratified sampling to preserve the original class distribution across all subsets we decided to reduce the number of classes to improve models performance[3][4].

6.1.2 Feature Extraction

Table 2: Feature Extraction Methods

Method	**Brief Description**
Bag of Words (BoW) [5]	Represents text as word frequency counts using unigrams, without considering word order.
Bag of N-grams [6]	Extracts sequences of words, such as bigrams, to capture short contextual patterns in the text.
TF-IDF [7]	Converts text into numerical features by giving higher weights to important words and lower weights to common words.
Feature Selection	Selects the top 300 most relevant TF-IDF features using the Chi-square statistical test.

6.1.3 Model Training and Validation

The dataset was divided into 15% testing, 15% validation and 70% training. After that, 3 models of machine learning were trained as follows:

Model 1: Logistic Regression
Considered as baseline to compare with later models.

Model 2: Random Forest
uses the RandomForestClassifier with 100 decision trees. This model builds multiple decision trees and combines their predictions to improve classification performance.

Model 3: Support Vector Machine (SVM)
uses the SVC classifier with a linear kernel. It is commonly used in text classification tasks because it performs well with high-dimensional textual features.

The same feature extraction methods were applied to all models to ensure a fair comparison. The model performance was evaluated using Accuracy, Precision, Recall, and F1-score on the validation set.

6.2 LLM-Based Approach

The modren approch is using LLM which refers to Transformer language models that contain hundreds of billions (or more) of parameters, which are trained on massive text data. LLMs exhibit strong capacities to understand natural language and solve complex tasks (via text generation).[8]

Unlike traditional approaches that use hand crafted feature extraction, the LLM approach rely on deep contextual understanding. In this project, we used the **ALLaM** model using a prompt-based approach to guide the model's output without any weight update.

6.2.1 Prompting Strategy

Table 3: Types of Prompting

Approach	**Brief Description**
Zero Shot	no examples, the model must rely on pretrained knowledge only to solve the task.
One Shot	one example are provided to clarify the task to the model, with strong rely on pretrained knowledge.
Few Shot	few examples are provided to help the model capture the pattern and adapt the task.

we applied all three approaches to experiment the effect of shots in our dataset context and to find out the model's robustness when changing the context size and the information provided [9].

6.2.2 Prompt Design

we use the same prompt to analyse the real impact of shots provided.

The Prompt:

You are an expert in Arabic sentiment analysis.
Task: Classify the following Arabic tweet into exactly one of these two categories: (Positive, Negative).
Strict Rules:
- You must answer with exactly ONE English word from the given options.
- Do NOT use the word 'Neutral'.
- Do NOT provide any explanations, notes, or additional text.

Zero Shot:
Considered as a baseline, provide the prompt and target tweet immediately.

One Shot:
Single example selected from outside the dataset with positive sentiment.

Few Shots (4-Shots):
provided 2 positive and 2 negative examples to prevent majority label bais. these examples selected from the dataset itself to help the model capture the dataset's structure and format.

6.3 Overall Pipeline

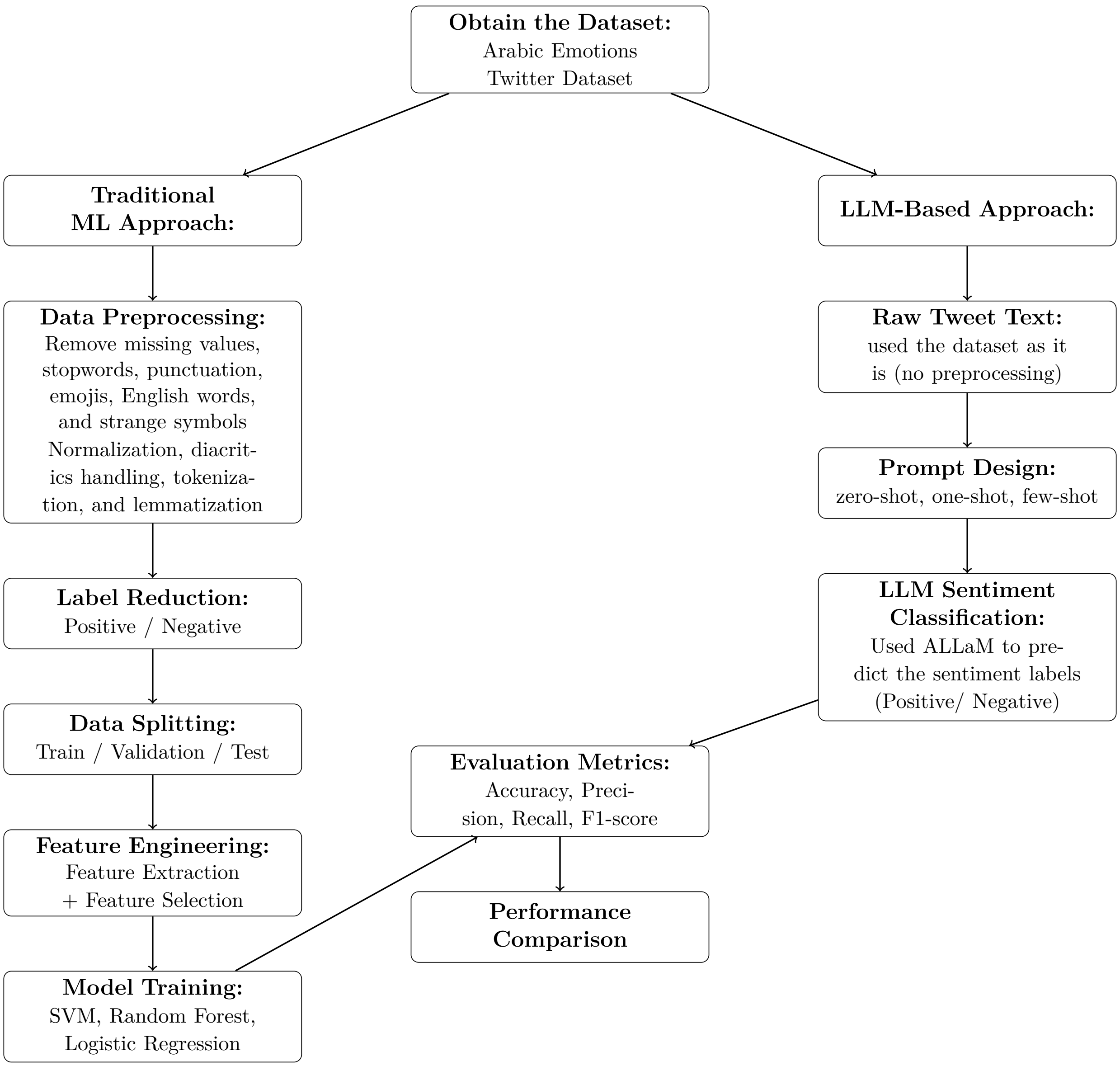

Figure 1: Overall pipeline of the traditional machine learning and LLM-based approaches.

7 Results

7.1 Traditional Approach Results

7.1.1 Validation Results and Model Selection

Table 4: Validation Results for Different Feature Methods and Models

Feature Method	Model	Accuracy	F1-score	Precision	Recall
TF-IDF	SVM	0.8536	0.8535	0.8536	0.8536
BoW	Logistic Regression	0.8312	0.8313	0.8317	0.8312
TF-IDF	Logistic Regression	0.8268	0.8264	0.8273	0.8268
BoW	SVM	0.8107	0.8108	0.8118	0.8107
TF-IDF	Random Forest	0.8089	0.8090	0.8101	0.8089
BoW	Random Forest	0.8062	0.8062	0.8062	0.8062
Feature Selection	Logistic Regression	0.8009	0.8000	0.8029	0.8009
Feature Selection	SVM	0.7938	0.7917	0.7996	0.7938
Feature Selection	Random Forest	0.7705	0.7703	0.7705	0.7705
N-grams	SVM	0.7438	0.7392	0.7536	0.7438
N-grams	Logistic Regression	0.7429	0.7332	0.7702	0.7429
N-grams	Random Forest	0.6214	0.5932	0.6923	0.6214

Based on the validation results, the SVM model with TF-IDF features achieved the highest performance compared to the other models. For this reason, it was selected as the final model for the testing phase.

7.1.2 Final Testing

After selecting the best model from the validation results, the final model was tested on the unseen test set. The selected model was SVM with TF-IDF features.

Table 5: Final Testing Results for the Selected Model

Metric	Score
Accuracy	0.8492
F1-score	0.8489
Precision	0.8501
Recall	0.8492

Table 6: Classification Report for the Final Model

Class	Precision	Recall	F1-score	Support
Negative	0.84	0.88	0.86	586
Positive	0.86	0.81	0.84	535
Macro Avg	0.85	0.85	0.85	1121
Weighted Avg	0.85	0.85	0.85	1121

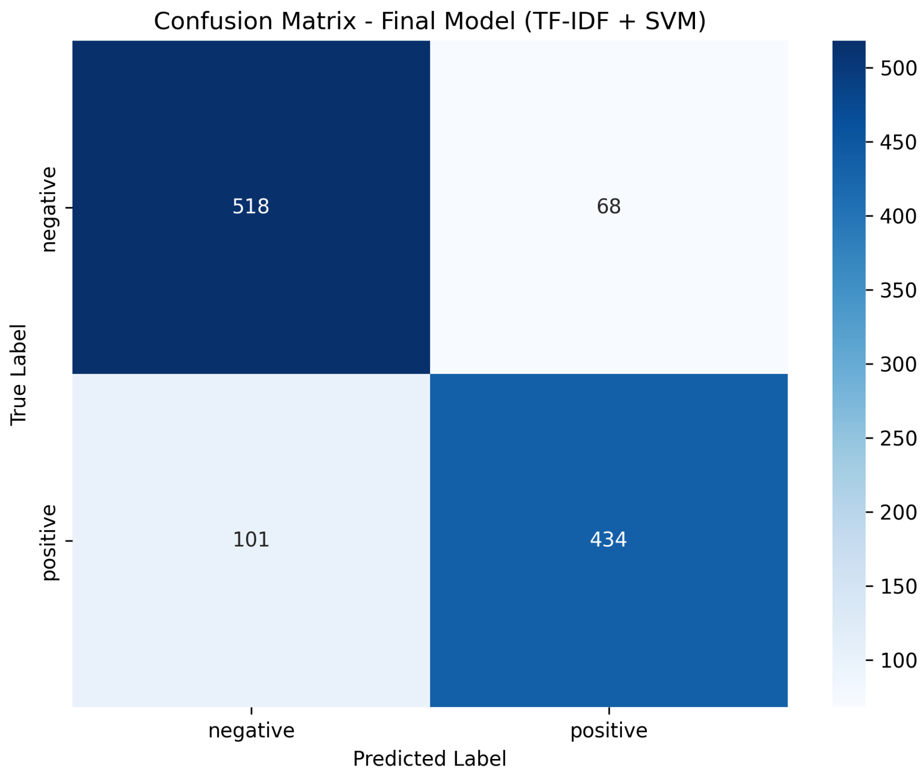

Figure 2: Confusion Matrix for the Final Model

7.2 LLM-Based Approach Results

Table 7: Results for Prompting Approaches

Approach	Accuracy	Precision	Recall	F1-score	Support
Zero Shot	0.85	0.86	0.85	0.86	500
One Shot	0.85	0.86	0.86	0.86	500
Few Shot	0.84	0.85	0.85	0.85	500

Based on the results, the One Shot approach achieved the highest performance, followed by Zero Shot approach. its appeared that ALLaM model had strong pretrained knowledge of Arabic Knowledge to perform the task well without needing to examples. The single example in One Shot was enough to guide the model. The lower performance of Few Shots approach indicates that more examples not necessarily improve the performance, sometimes it's just adding noise and make it unstable. [10]

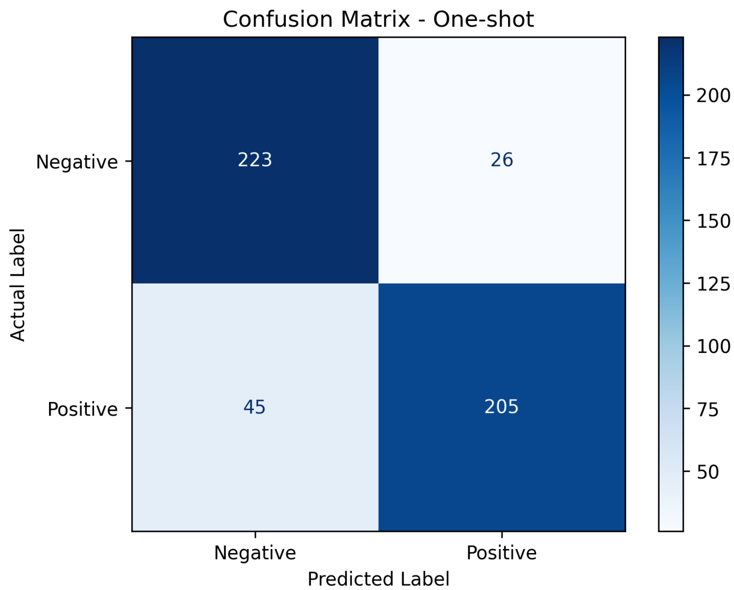

Figure 3: Confusion matrix for the One Shot prompting approach.

One Shot Prompt achieved the highest performance compared to the other approaches. For this reason, it was selected as the final model for the comparison phase.

7.3 Comparison Between Approaches

Table 8: Traditional vs LLM Comparison

Metric	**TF-IDF + SVM**	**One Shot Prompt**
Accuracy	0.84	0.85
F1-score	0.84	0.86
Precision	0.85	0.86
Recall	0.84	0.86

One shot prompting achieved slightly higher performance compared to TF-IDF + SVM with approximately 1-2% improvement across all metrics.

ALLaM (LLM) uses deep contextual embedding, allowing it to understand the meaning and grasp the context of sentiment tweets. it didn't need label data, no training and no weights update, making it very flexible and adapting to new requirements. However a key limitation that constrains ALLaM performance from achieving higher results is diversity of Arabic dialects contained in the dataset.

On the other hand, the traditional approach relies only on token frequency and matching keywords making it dependent on training vocabulary without any contextual understanding.

However it's reached close results to LLM at a lower computational cost, due to simple nature of sentiment analysis which often have some words that frequently appear to express some emotions. this making it a practical choice when resources are limited.

8 Discussion

Although large language models are expected to achieve strong performance in classifying Arabic sentiments, the results showed that the traditional SVM model using TF-IDF also achieved very close performance, and this raises a point of discussion, which is: **How is the traditional machine learning model still able to compete with methods based on large language models in this task?** One possible explanation is that simplifying the task into a binary classification made the differences between the classes clear. Which allowed traditional machine learning models to perform efficiently

9 Conclusion

This project reviewed and compared traditional machine learning approach and LLMs based apporach in classifying arabic sentiments. The results showed that both approaches achieved strong and similar performance, which confirms the importance of evaluating both traditional and LLMs approaches rather than assuming that LLMs will always provide significantly better results

References

[1] A. Al-Khatib and S. R. El-Beltagy, “Emotional tone detection in arabic tweets,” in *18th International Conference on Computational Linguistics and Intelligent Text Processing*, 2017.

[2] O’Reilly Media, “Natural language processing fundamentals,” 2018, accessed: 2026-05-11. [Online]. Available: https://www.oreilly.com/library/view/natural-language-processing/9781788478311/ch01s03.html

[3] Noura, “Python3: Remove arabic punctuation,” 2018, accessed: 2026-05-16. [Online]. Available: https://stackoverflow.com/questions/48496869/python3-remove-arabic-punctuation

[4] CAMeL Lab, “Camel tools documentation,” https://camel-tools.readthedocs.io/en/latest/, 2024, accessed: 2026-05-16.

[5] DataCamp, “Bag of words model in python,” 2024, accessed: 2026-05-11. [Online]. Available: https://www.datacamp.com/tutorial/python-bag-of-words-model

[6] eCampusOntario, “mplementing and analyzing n-grams in python,” 2023, accessed: 2026-05-15. [Online]. Available: https://ecampusontario.pressbooks.pub/nudh3/chapter/implementing-and-analyzing-n-grams-in-python/

[7] GeeksforGeeks, “Understanding tf-idf (term frequency-inverse document frequency),” 2024, accessed: 2026-05-11. [Online]. Available: https://www.geeksforgeeks.org/machine-learning/understanding-tf-idf-term-frequency-inverse-document-frequency/

[8] Wayne Xin Zhao, Kun Zhou, Junyi Li, Tianyi Tang, Xiaolei Wang, Yupeng Hou, Yingqian Min, Beichen Zhang, Junjie Zhang, Zican Dong, Yifan Du, Chen Yang, Yushuo Chen, Zhipeng Chen, Jinhao Jiang, Ruiyang Ren, Yifan Li, Xinyu Tang, Zikang Liu, Peiyu Liu, Jian-Yun Nie and Ji-Rong Wen, “A survey of large language models,” 2023. [Online]. Available: https://arxiv.org/abs/2303.18223

[9] Prompting Guide, “Prompt engineering techniques,” 2024, accessed: 2026-05-16. [Online]. Available: https://www.promptingguide.ai/techniques

[10] Yao Lu, Max Bartolo, Alastair Moore, Sebastian Riedel, Pontus Stenetorp, “Fantastically ordered prompts and where to find them: Overcoming few-shot prompt order sensitivity,” 2022. [Online]. Available: https://arxiv.org/abs/2104.08786